

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\alpha_k := \frac{2}{k+1}, \beta_k := \frac{1}{2L_{Ef}}, \lambda_k := \frac{k\beta_k}{2}, m_k := \left\lceil \frac{L_{Ef}k}{\tilde{D}^2} \right\rceil, \text{ and}$$

$$\Pr(R=k) := \frac{\Gamma_k^{-1}\beta_k(1-L_{Ef}\beta_k)}{\sum_{\tau=1}^N \Gamma_{\tau}^{-1}\beta_{\tau}(1-L_{Ef}\beta_{\tau})}, \text{ for } k=1,2,\dots,N,$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} ((Q_0 \cdot Q_{A_{\psi_0}})(\iota(\pi))|\mathcal{L}_{\psi_0}) &= 0, \\ \text{beginalign*2mm}] \bigwedge^2(Q_{A_{\psi_0}}(\iota(\pi))|\mathcal{L}_{\psi_0}) &= 0, \\ \text{beginalign*2mm}](Q_{A_{\psi}}(\iota(\pi))|\mathcal{L}_{\psi}) &= 0, \quad \psi \neq \psi_0. \\ \text{beginalign*3mm}] & \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} (\lambda C_{\lambda})^{(p-1-\alpha)(p-1-k_2)-k_1k_3} &= \left(\frac{c_1}{\mu^{k_1}B_{\lambda}} \right)^{p-1-k_2} \left(\frac{c_2}{\mu^{k_2}B_{\mu}} \right)^{k_1}, \\ (\mu C_{\mu})^{(p-1-\alpha)(p-1-k_2)-k_1k_3} &= \left(\frac{c_2}{\mu^{k_2}B_{\mu}} \right)^{p-1-\alpha} \left(\frac{c_1}{\mu^{k_1}B_{\lambda}} \right)^{k_3}. \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} S_{\text{off}}^{c < p^2} &:= \sum_{d_1, d_2 < D} \sum_{Id_1, Jd_2 < D} x_{Id_1} \bar{x}_{Jd_2} \sqrt{IJ} \sum_{L, M \geq 1} \frac{G_k(LMd_1d_2/p)}{\sqrt{LM}} \\ &\quad \times \left(2\pi i^{2k} \sum_{\substack{p|c \\ c < p^2}} \frac{\mathcal{S}(IL, JM, c)}{c} J_{2k-1} \left(\frac{4\pi\sqrt{ILJM}}{c} \right) \right) \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$u_i^{35,35}(y, z) = \left\{ \begin{array}{ll} -5b_i^{35,35}(y, z) & if \\ -1 & if \\ 1 & if \end{array} \begin{array}{l} |5b_i^{35,35}(y, z)| \leq 1, \\ -5b_i^{35,35}(y, z) < -1, \\ 5b_i^{35,35}(y, z) > 1, \end{array} \right\}, \quad i = 1, 2,$$